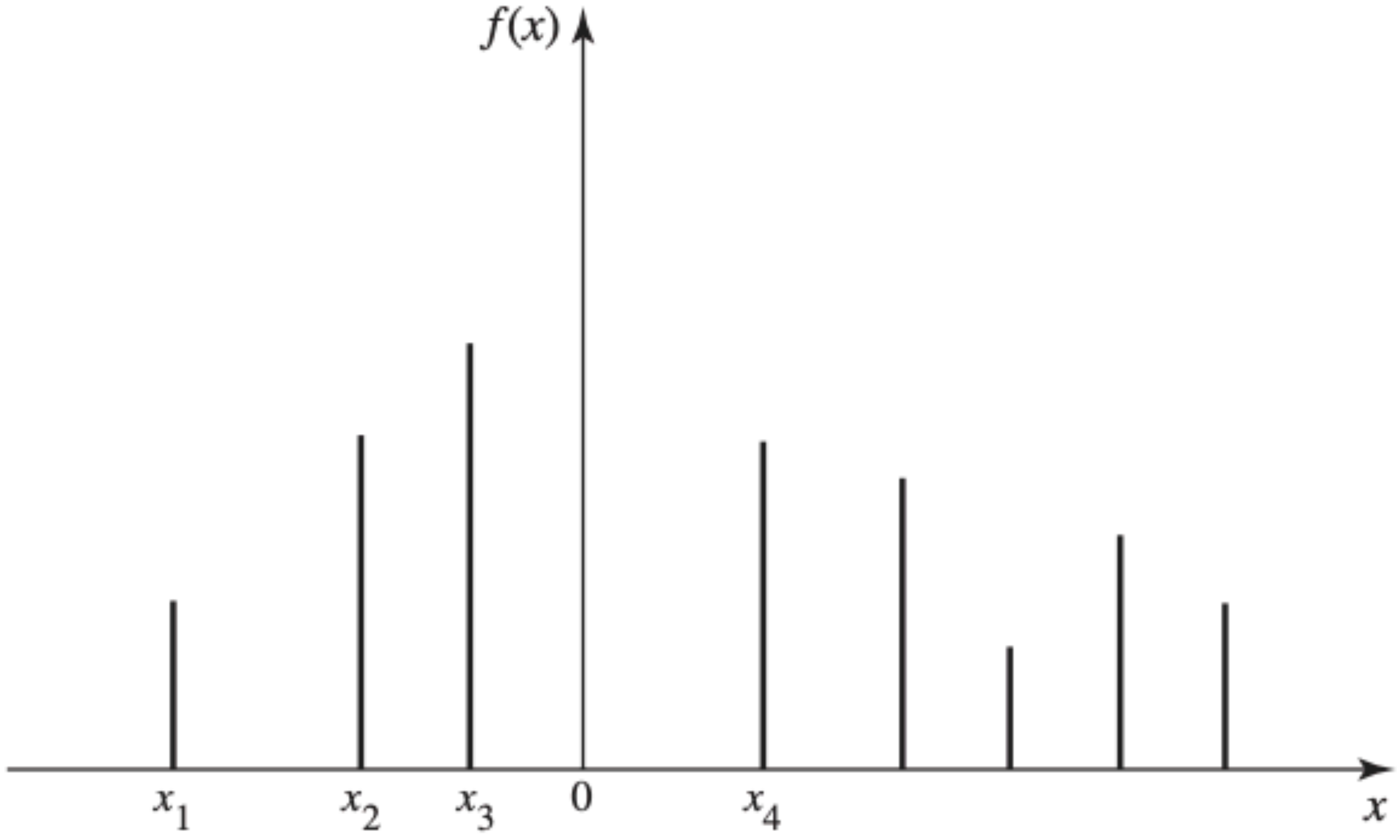
QUIZ 1

What is the probability of x heads in n tosses of a fair coin?

(x can be any one of $0, 1, \cdots n$)

REVIEW OF (LAST PART OF) LECTURE 3



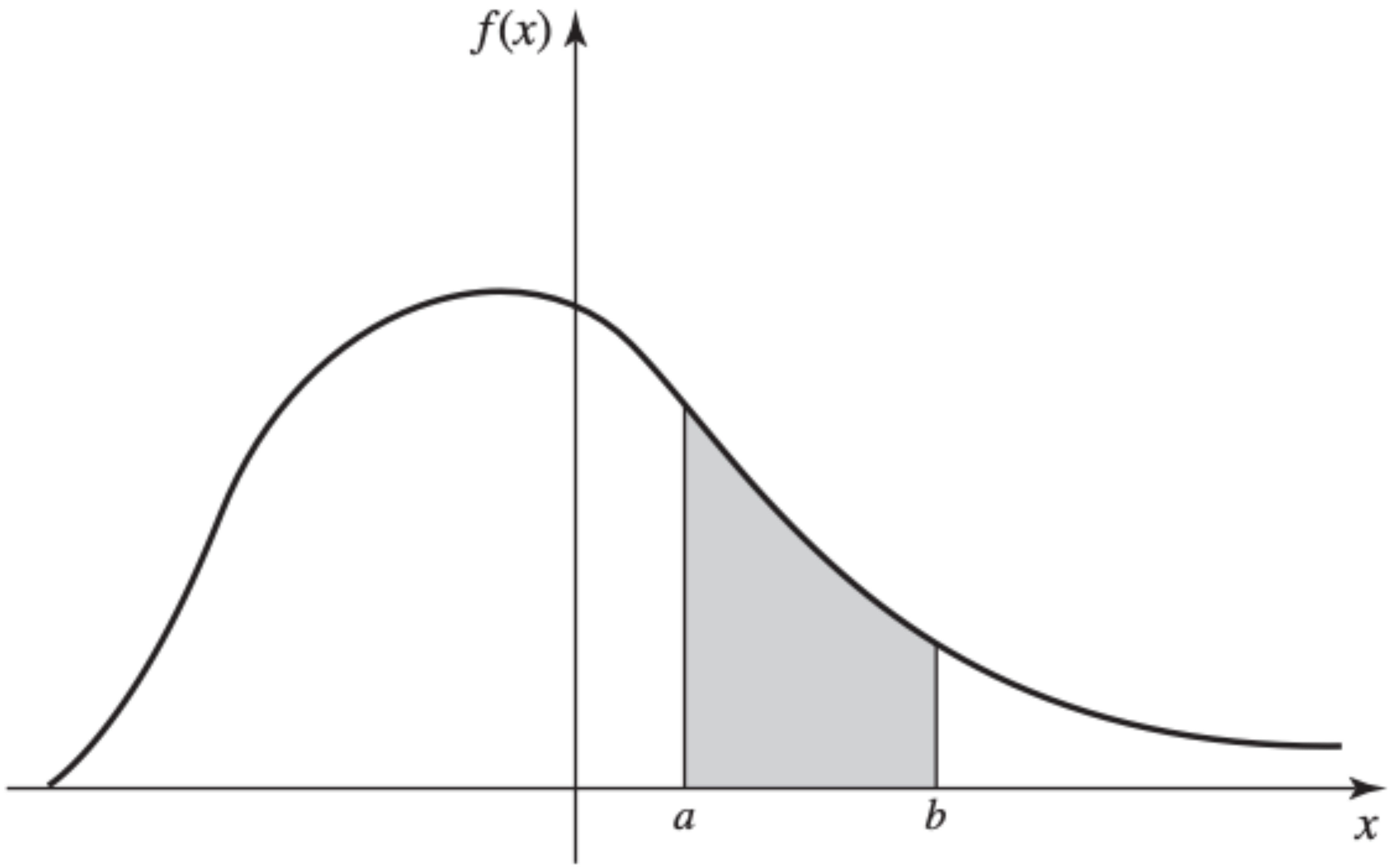
- pmf of a discrete random variable X:

$$f(x) = P(X = x)$$

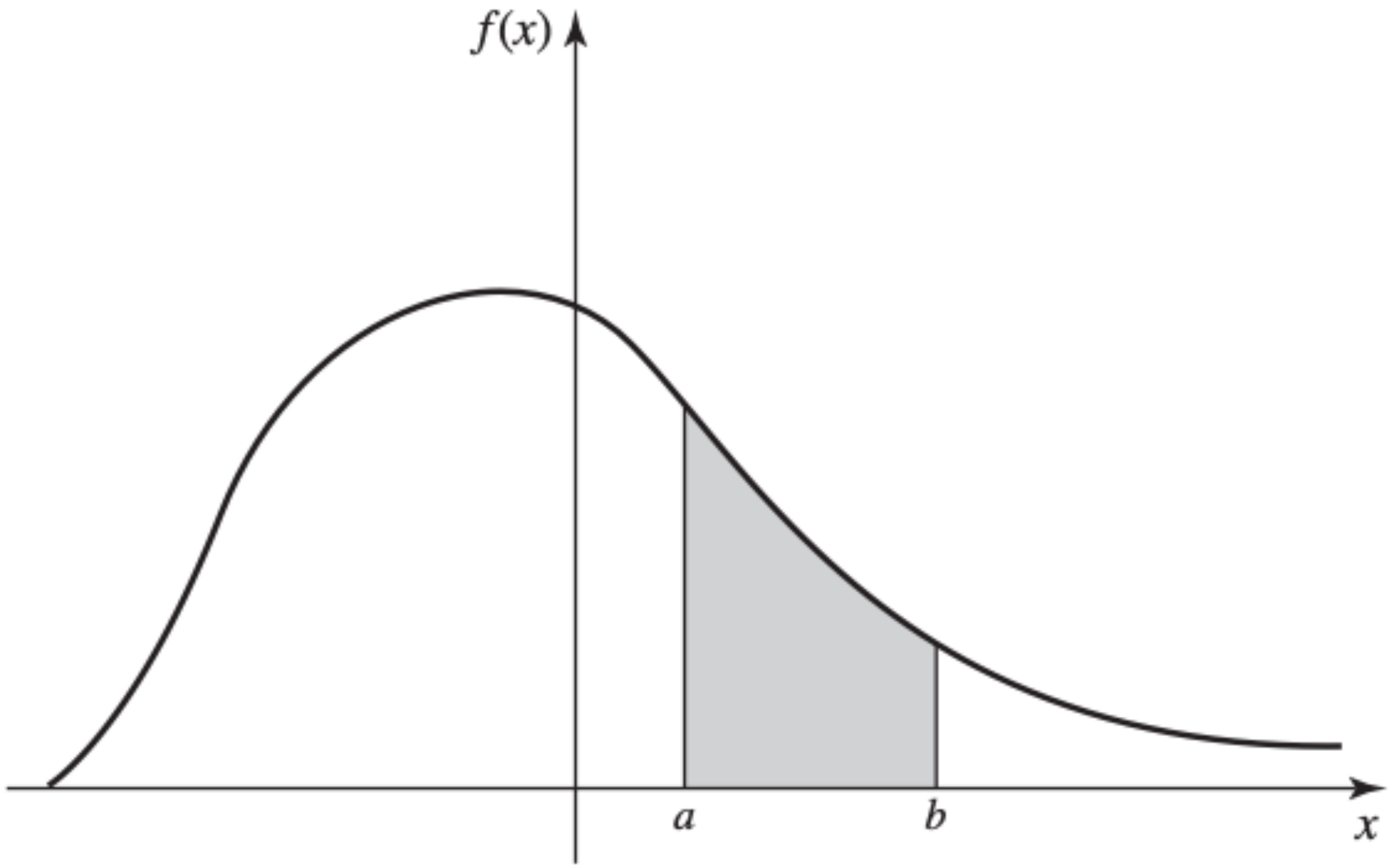
- For any pmf:
  - $f(x) \geq 0$
  - $\sum_{x} f(x) = 1$

- For an interval: $P(x_2 \leq X \leq x_4) = \sum_{x \in \{x_2, x_3, x_4\}} f(x)$

CONTINUOUS DISTRIBUTIONS AND RANDOM VARIABLES



CONTINUOUS DISTRIBUTIONS AND RANDOM VARIABLES



- If $\exists f$ s.t. $\forall [a, b]$ s.t. $a, b \in \mathbb{R}$,

$$P(a \leq X \leq b) = \int_a^b f(x)\, dx$$

$\Rightarrow X$ continuous random variable

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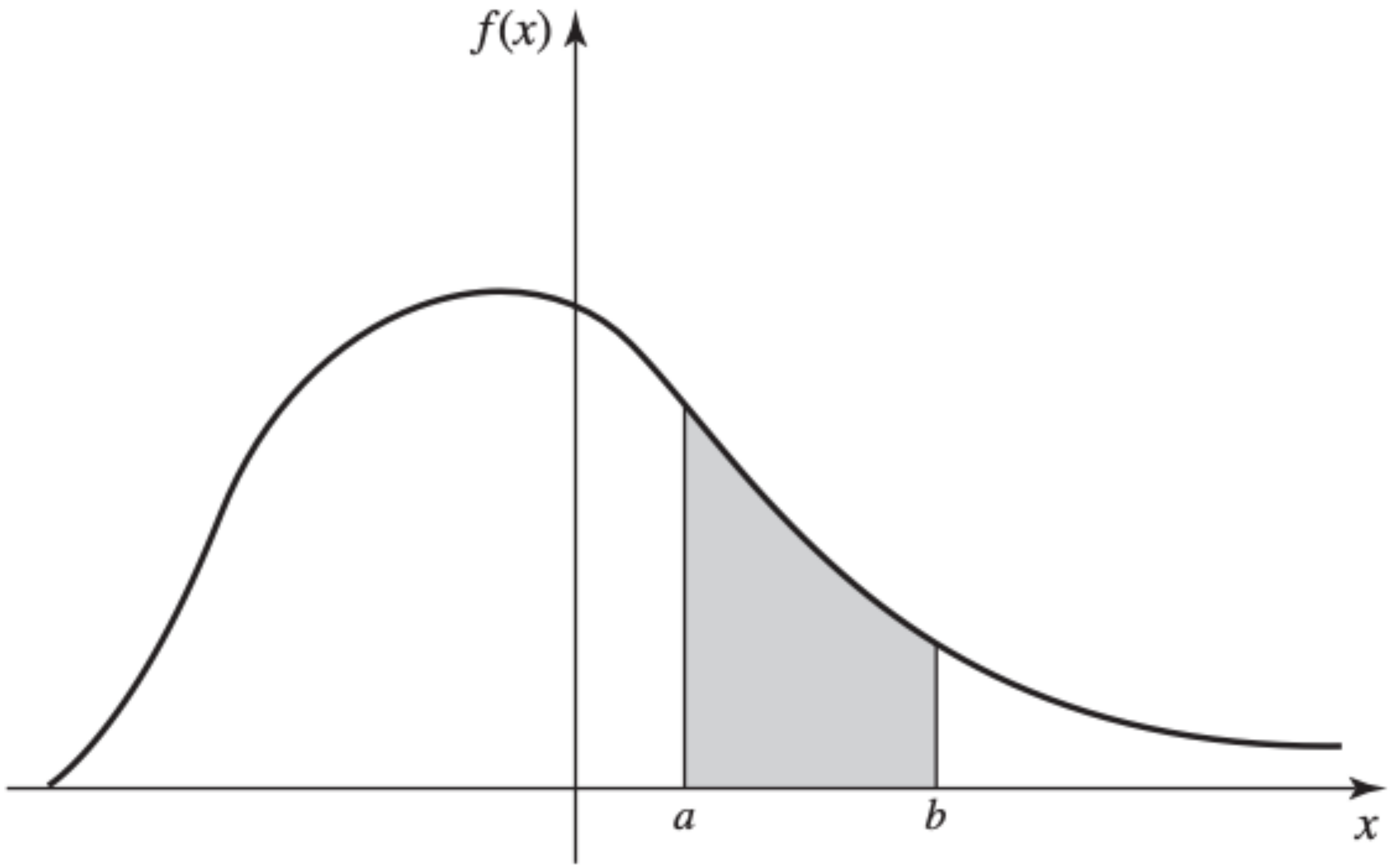

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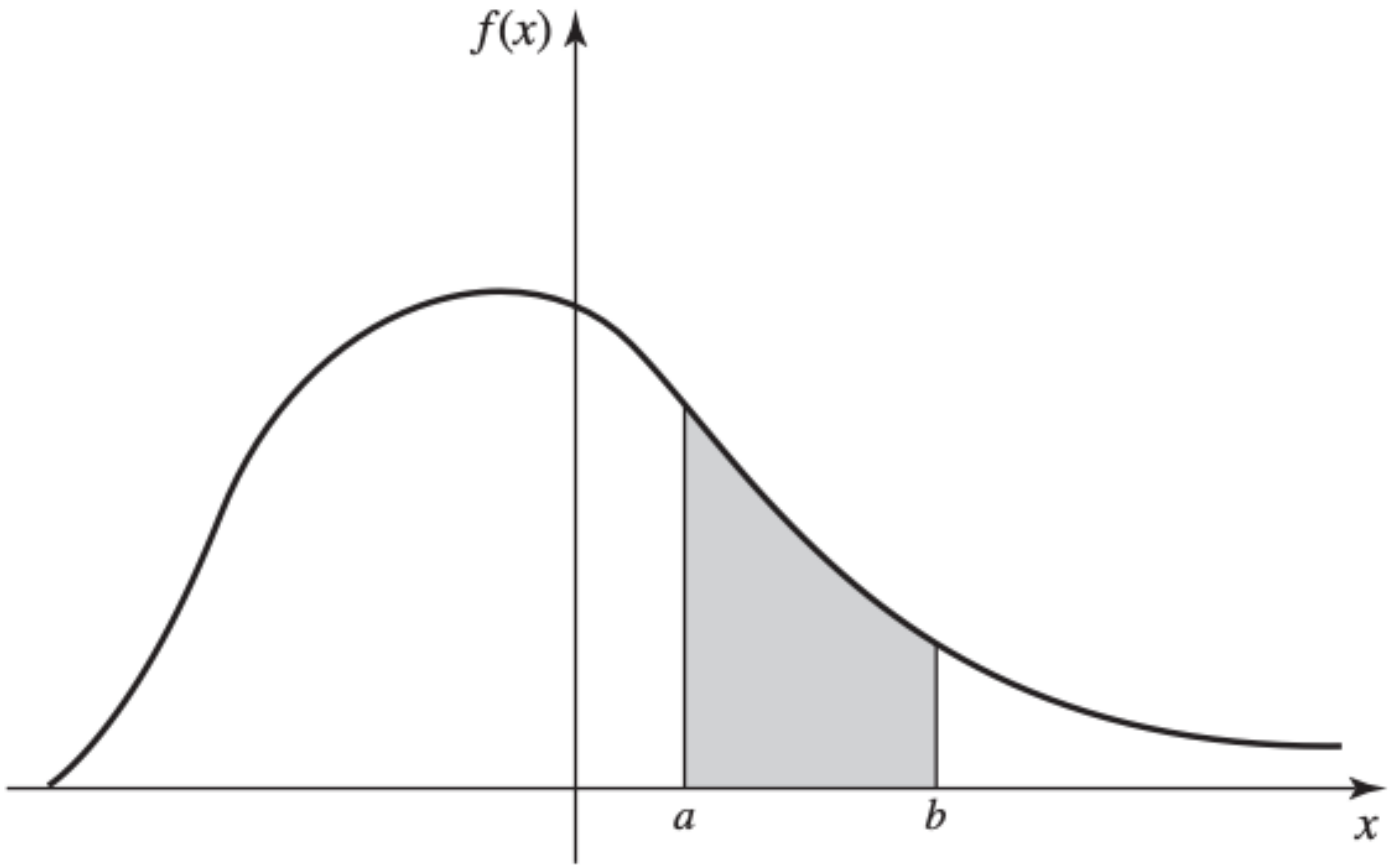
Reminder: $P(a \leq X \leq b) = \sum_{x=a}^{b} f(x)$ for discrete RV X.

CONTINUOUS DISTRIBUTIONS AND RANDOM VARIABLES



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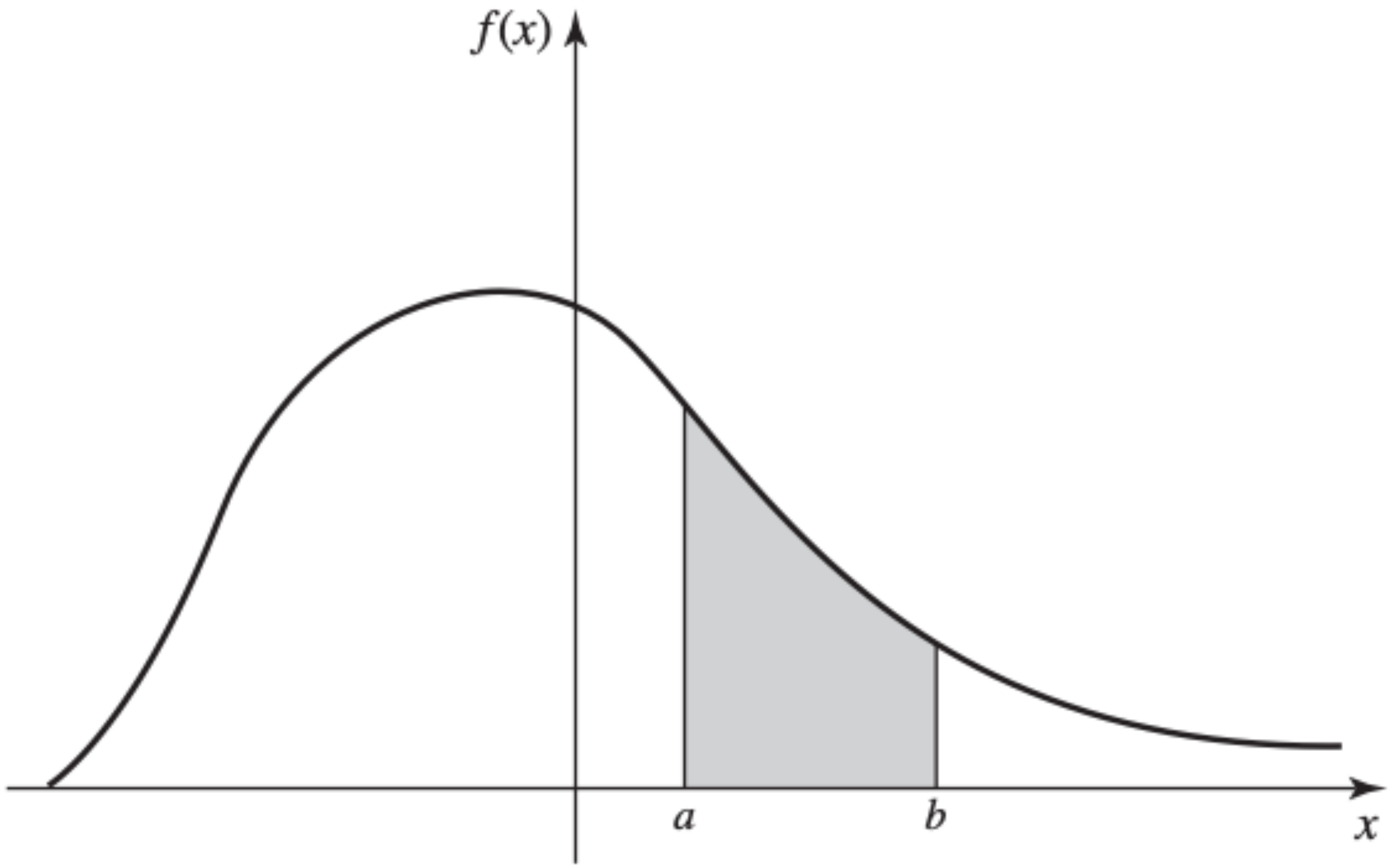
- $f(x)$ is the probability density function (pdf) of X.
- $\{x : f(x) > 0\}$ is the support of X.

CONTINUOUS DISTRIBUTIONS AND RANDOM VARIABLES



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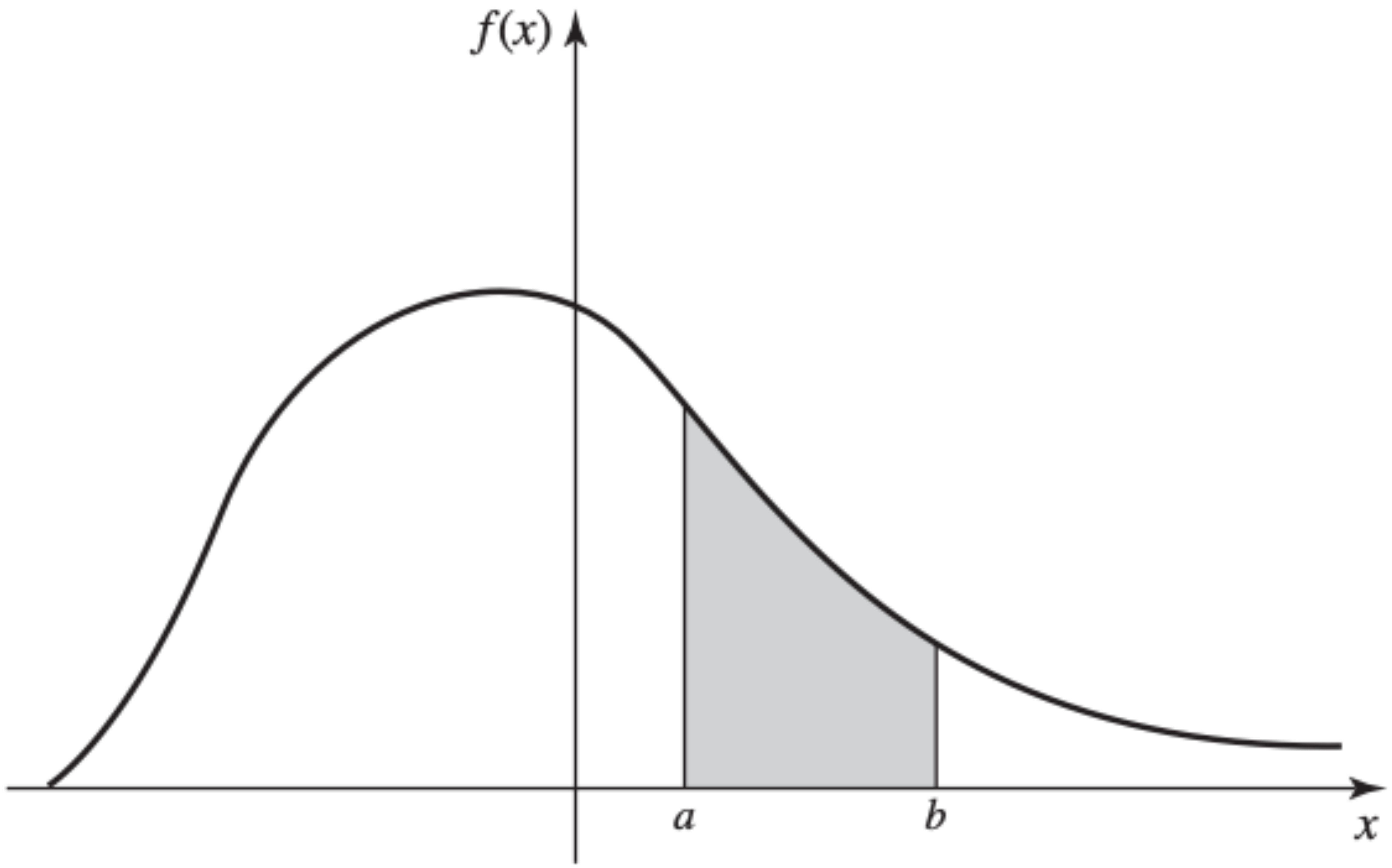
Note: $P(x = a) = 0, \ \ \forall a.$

CONTINUOUS DISTRIBUTIONS AND RANDOM VARIABLES



- For any pdf:
  - $f(x) \geq 0, \quad \forall x$
  - $\int_{-\infty}^{\infty} f(x) \; dx = 1$

CONTINUOUS DISTRIBUTIONS AND RANDOM VARIABLES

Ex: $a \leq X \leq b$ and $\forall$ subinterval, probability X in it is proportional to the length of subinterval

$\Rightarrow X$ has uniform distribution on $[a, b]$. What is the pdf of X?

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Ex: $a \leq X \leq b$ and $\forall$ subinterval, probability X in it is proportional to the length of subinterval

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Probability that X is in any interval with same length is the same

$\Rightarrow f(x)$ is some constant c throughout $[a, b]$. Furthermore,

$$\int_{-\infty}^{\infty} f(x)\, dx = \int_{a}^{b} c\, dx = 1 \;\Rightarrow\; f(x) = \begin{cases} \dfrac{1}{b-a} & \text{for } a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$$

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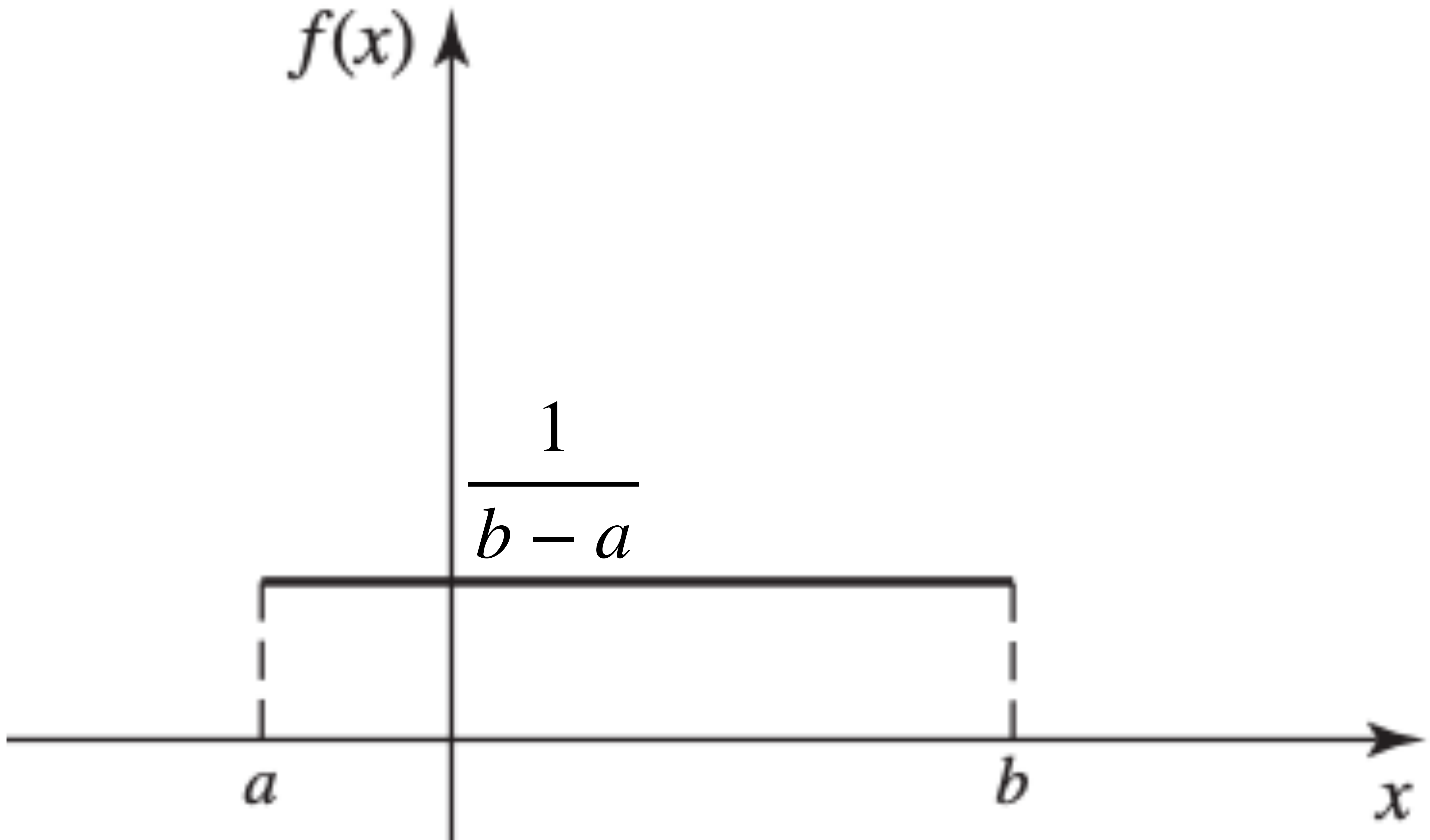

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Note: Density is not probability.

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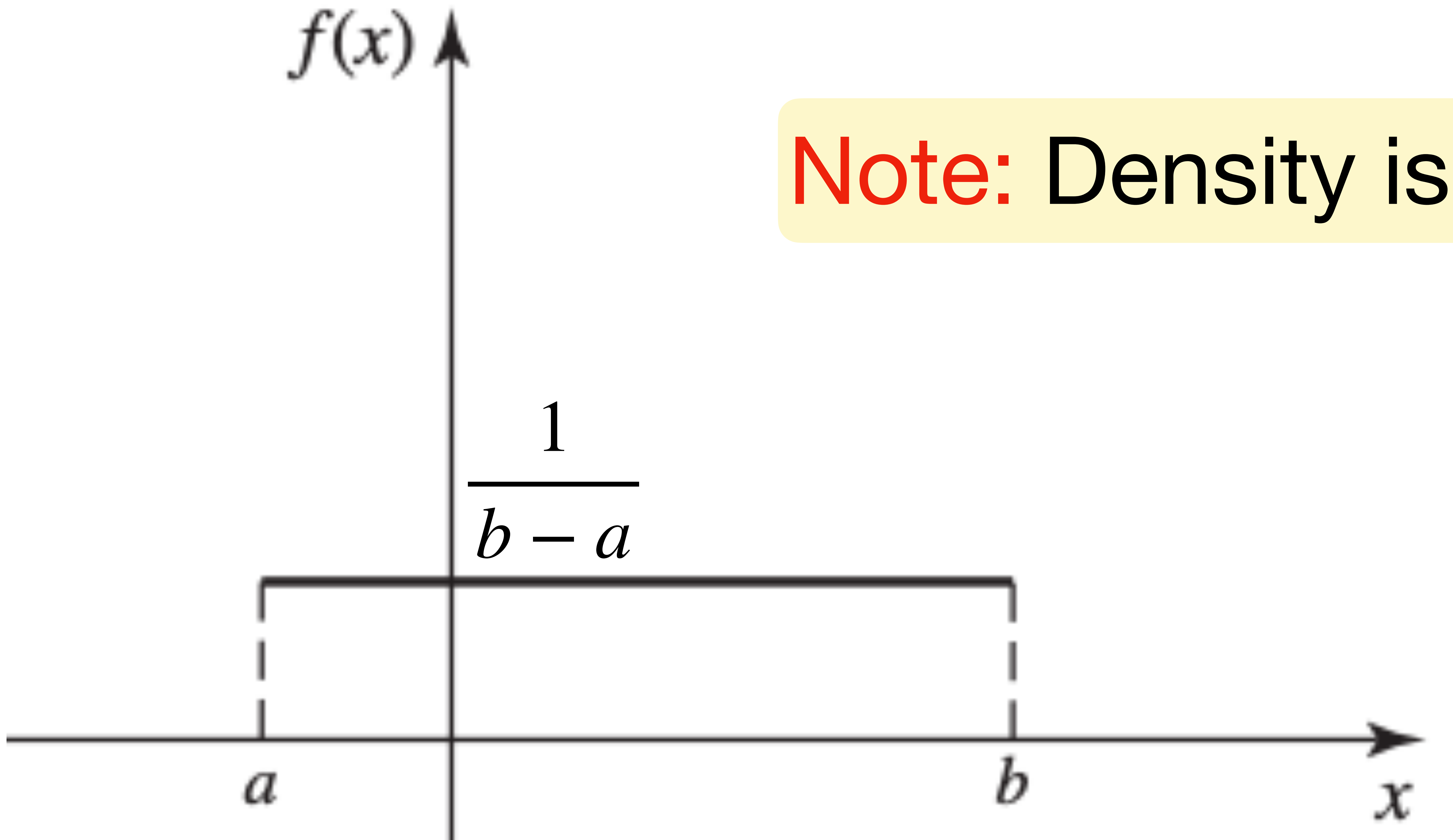

CUMULATIVE DISTRIBUTION FUNCTION

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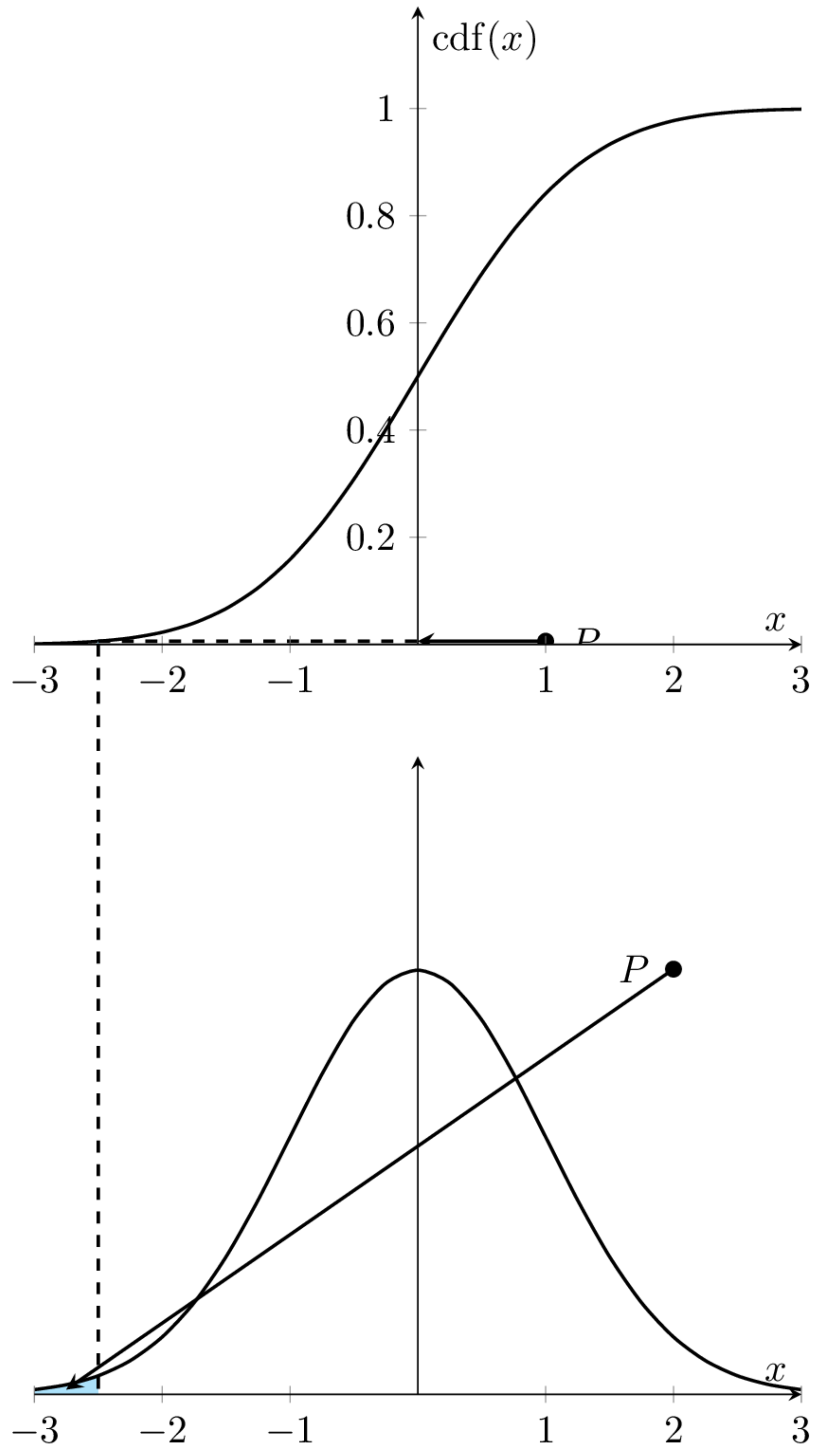
- $P(X > x) = 1 - F(x)$ and $P(a < X \leq b) = F(b) - F(a)$

CUMULATIVE DISTRIBUTION FUNCTION

- Relationship between cdf and pdf

$$F(x) = P(X \leq x) = \int_{-\infty}^{x} f(t)\, dt$$

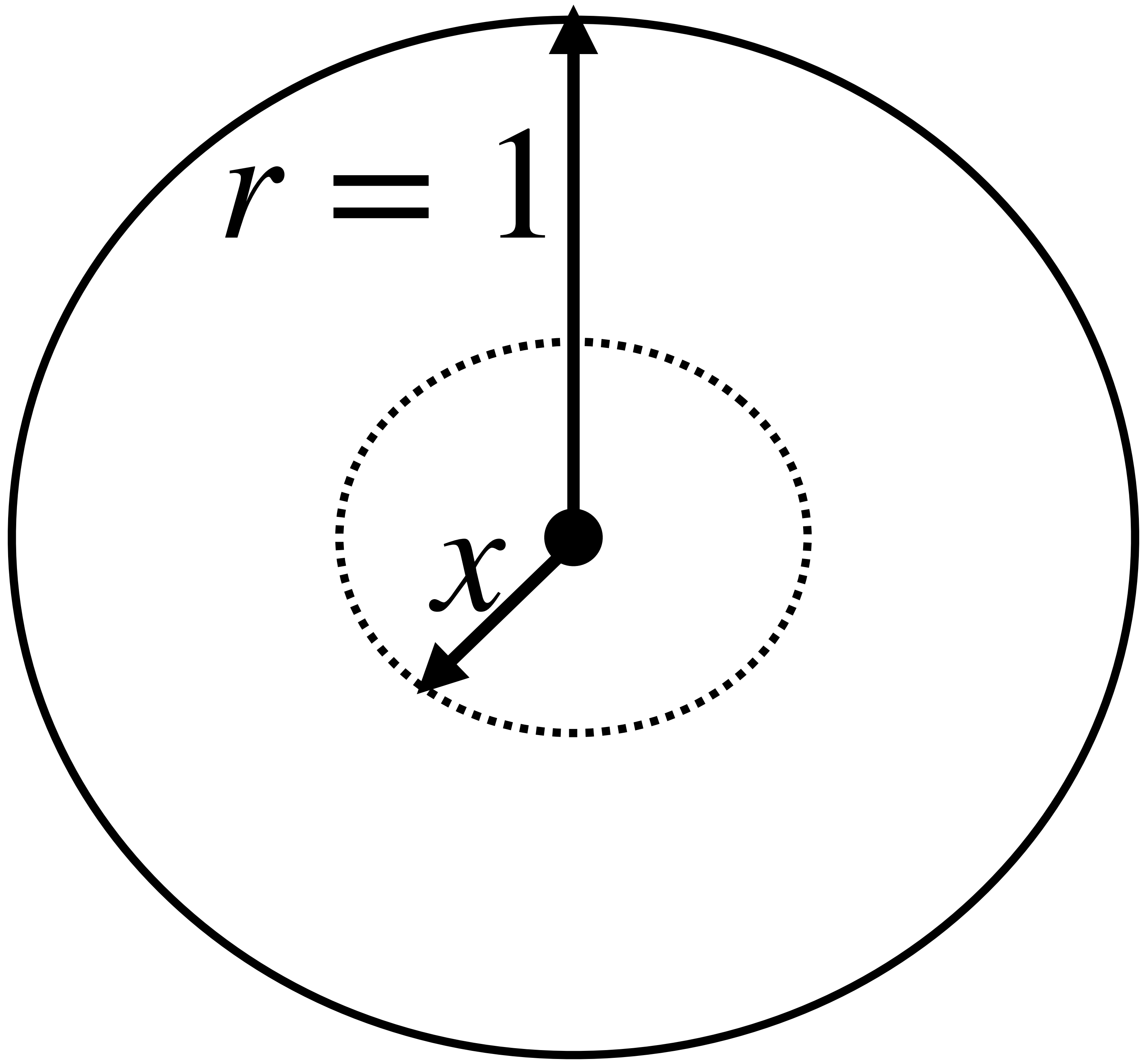


cdf $F(x)$

pdf $f(x)$

CUMULATIVE DISTRIBUTION FUNCTION

Ex: Dart with radius 1. Lands randomly uniformly on the board.



X : distance from the center

What is the cdf of X, that is $F(x)$?

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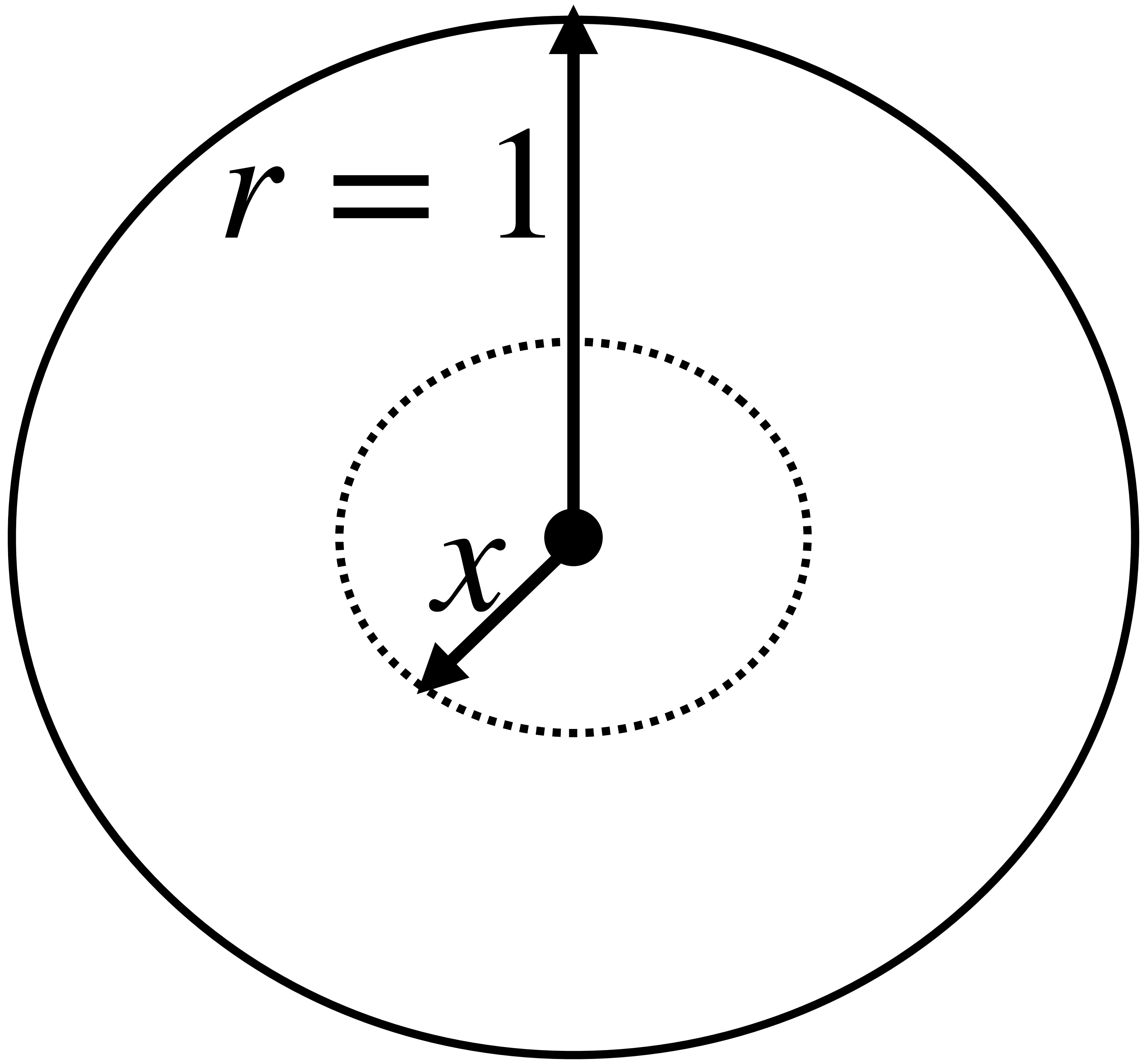

X : distance from the center

What is the cdf of X, that is $F(x)$?

$$P(X \leq x) = \frac{\pi x^2}{\pi 1^2} = x^2 \quad \Rightarrow$$

$$F(x) = \begin{cases} 0 & \text{if } x \leq 0 \\ x^2 & \text{if } 0 < x \leq 1 \\ 1 & \text{if } 1 < x \end{cases}$$

CUMULATIVE DISTRIBUTION FUNCTION

Ex: If X has uniform distribution on $[a, b]$, we found the pdf:

$$f(x) = \begin{cases} \dfrac{1}{b-a} & \text{for } a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$$

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What is the cdf of X?

$$F(x) = \begin{cases} 0 & \text{if } x \leq a \\ \displaystyle\int_a^x \frac{1}{b-a}\, dt & \text{if } a < x \leq b \\ 1 & \text{if} x > b \end{cases} \quad \Rightarrow \quad F(x) = \frac{x-a}{b-a}, \ \forall a < x \leq b$$

JOINT DISTRIBUTIONS: DISCRETE

- The joint pmf of discrete random variables X, Y:

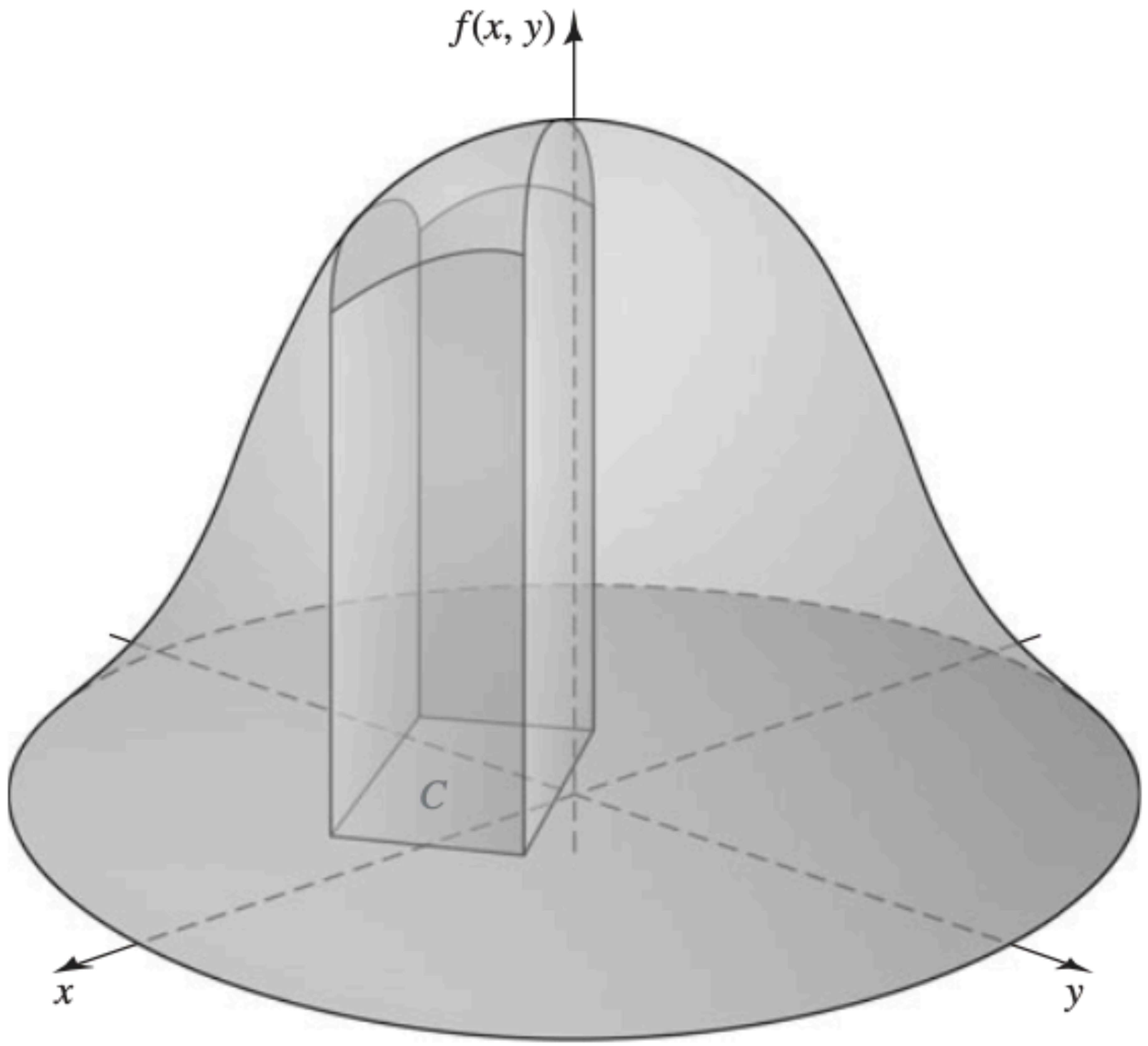
$$f(x, y) = P(X = x \text{ and } Y = y)$$

Note that $\displaystyle\sum_{\forall (x,y)} f(x, y) = 1.$

JOINT DISTRIBUTIONS: CONTINUOUS

- A joint pdf satisfies:

$$f(x, y) \geq 0 \quad \text{AND} \quad \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y)\, dx\, dy = 1$$



$$P((X, Y) \in C) = \int_C \int f(x, y)\, dx\, dy$$

JOINT DISTRIBUTIONS: DISCRETE

Ex: X: # of cars owned by a randomly selected household

Y: # of computers owned by the same household

Joint pmf shown with a table. For our example:

	y			
x	1	2	3	4
1	0.1	0	0.1	0
2	0.3	0	0.1	0.2
3	0	0.2	0	0

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Probability that a randomly selected household has ≥ 2 cars and computers?

Sum of values gives $P(X \geq 2 \text{ and } Y \geq 2)$

$= 0.5$

MARGINAL DISTRIBUTIONS: DISCRETE AND CONTINUOUS

- Given joint distribution of X, Y need distribution of one, say X. Such a distribution is marginal distribution of X.

  f: joint pmf of X, Y. Marginal pmf of X:

$$f_1(x) = \sum_{\forall y} f(x, y)$$

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  f: joint pmf of X, Y. Marginal pmf of X:

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- For the continuous case:

  f: joint pdf of X, Y. Marginal pdf of X:

$$f_1(x) = \int_{-\infty}^{\infty} f(x, y)\, dy$$

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Joint pmf shown with the table.

	y				
x	1	2	3	4	Total
1	0.1	0	0.1	0	0.2
2	0.3	0	0.1	0.2	0.6
3	0	0.2	0	0	0.2
Total	0.4	0.2	0.2	0.2	1.0

Marginal pmf of X: $f_1(X)$

Marginal pmf of Y: $f_2(Y)$